Deliverable One: Leadership Interview

Task D1.1: Pizzeria Owner/Manager Identification/Visit:

For this project, our team chose to work with a small, independently owned neighborhood pizzeria managed directly by its manager, Lev. During our first visit, we introduced ourselves as graduate students and explained that the purpose of our project was to explore the financial decision-making processes of small businesses. By approaching the interaction with professionalism and genuine curiosity, we were able to establish a comfortable dialogue. Lev responded warmly, sharing his enthusiasm for both his business and our academic goals. This mutual interest created an immediate sense of trust and respect. By the end of the meeting, Lev not only welcomed our initiative but also encouraged us to continue our study and invited us for a follow-up discussion. This made our initial visit both engaging and highly productive, laying a strong foundation for the project.

Task D1.2: Leadership Interview and Theory Questionnaire Preparation:

To prepare for the interview, our team designed a customized questionnaire that connected academic insight with the practical realities of managing a small business. Instead of generic prompts, we built a focused framework around three core theories, Agency Theory, Stakeholder Theory, and Information Asymmetry, to encourage deeper reflection and tie theoretical concepts to the manager's real-world experiences.

Agency Theory: We began by examining the potential misalignment between an manager's objectives and employee actions. To explore this, we designed focused questions such as: *"Have you ever encountered situations where employees acted in ways that did not align with your business goals?"* and *"What systems or practices do you use to ensure that staff decisions support the long-term success of the pizzeria?"*

Stakeholder Theory: Next, we explored how the business balances the diverse interests of the groups it serves. Our custom questions included: *"Beyond your customers, which groups—such as suppliers, employees, or the local community, do you consider most critical to the pizzeria's success?"* and *"Can you share an example of when the needs of one stakeholder group conflicted with those of another, and how you navigated that situation?"*

Information Asymmetry: Finally, we investigated how uneven access to information can create challenges in decision-making. To address this, we prepared questions like: *"Do you believe that suppliers or distributors sometimes have more information than you about market prices or product quality?"* and *"What steps do you take to reduce information gaps and build trust with new employees, customers, or vendors?"*

By aligning each set of questions with a specific theory, we ensured that the interview would go beyond surface-level discussions. This structured yet flexible approach gave us the opportunity

to connect academic frameworks directly to the manager's real-world experiences, setting the stage for meaningful insights.

Task D1.3: Theory Selection:

Our discussion with the pizzeria manager highlighted two financial theories that were particularly relevant: Agency Theory and Information Asymmetry. Agency Theory came to life as we observed the manager's ongoing challenge of ensuring employees act in the business's best interest rather than prioritizing personal convenience. Likewise, Information Asymmetry became evident in the risks that arise when suppliers or distributors hold more knowledge about product quality or market pricing than the manager, creating potential vulnerabilities in decision-making.

Task D1.4: Active Experimentation Engagement:

The interview provided a practical lens through which we could see both theories at work in Lev's business.

- **Agency Theory:** Lev noted that while he relies on his staff to manage day-to-day operations, he sometimes worries they may not uphold the same standards of customer care that he values. To address this, he conducts regular check-ins and uses performance-based incentives to align employee actions with his own goals for the pizzeria.
- **Information Asymmetry:** Lev shared that, when working with new suppliers, he initially lacked clarity on fair market prices. To overcome this, he developed strategies such as comparing quotes across vendors and seeking referrals from other local businesses, allowing him to negotiate from a stronger position.

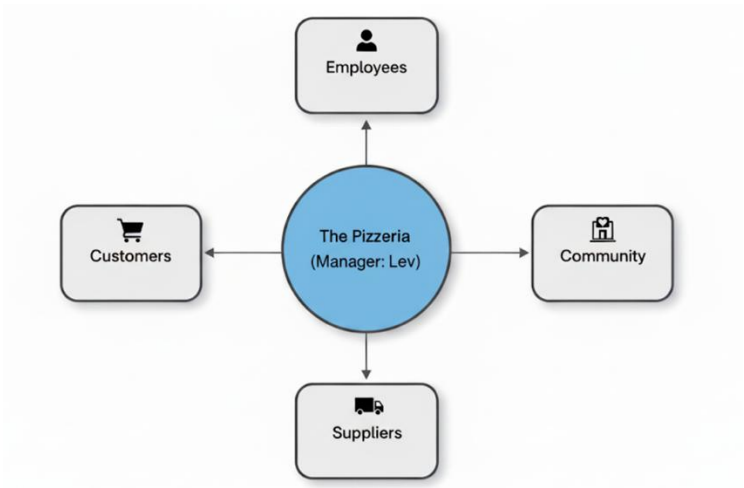

Figure 1: Key Stakeholders of Lev's Pizzeria

Task D1.5: Connecting Theory:

The interview with Lev played a crucial role in bridging academic theory with real-world application. Hearing directly from a business manager gave us theory-specific insights that transformed abstract concepts into practical realities.

- **Agency Theory:** Our understanding of this theory shifted from a textbook definition to the daily challenge of balancing trust and oversight. Lev's use of performance-based incentives and regular staff monitoring illustrated how small business managers actively manage agency conflicts to keep employee actions aligned with business goals.
- **Information Asymmetry:** Lev's experiences with supplier negotiations clarified this concept in a practical context. By comparing bids and seeking referrals, he demonstrated proactive strategies for reducing information gaps and protecting the business from unfavorable deals. This example showed how theory translates into concrete tools for risk management in small enterprises.

Task D1.6: Concrete Experience:

Our interview took place on-site at the pizzeria, an immersive setting that greatly enriched our learning. Sitting near the main counter, we experienced the aromas of fresh pizza and the constant movement of customers, turning the atmosphere into a live case study of small business operations.

Watching Lev shift seamlessly between serving customers and answering our questions highlighted the daily pressures of entrepreneurship. In just a few moments, he moved from hands-on service to strategic reflection, embodying the balancing act of a small business manager. This dynamic interaction made the theories we studied feel tangible and relatable, transforming abstract concepts into lived experience.

Figure 2: Our team outside Little Pizza Parlor

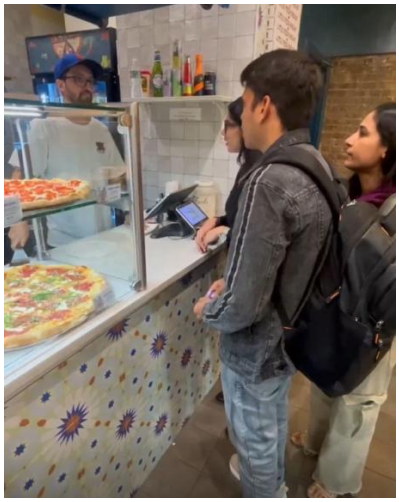

Figure 3: Interview

Task D1.7: Reflection:

The outcome of this project was significantly different from what I had anticipated. Standing in the pizzeria and conversing with Lev made it much more tangible than I had anticipated, as I had assumed it would primarily be about applying theories to a small business. I learned a lot that how much work goes into operating even a small local business by watching him handle orders, interact with customers, and still find time to answer our questions.
What struck me most was how the things we read in class actually show up in real life. For example, when Lev talked about dealing with suppliers or keeping employees motivated, I could immediately connect it back to concepts we've studied. It was surprising to see that what feels abstract in a textbook is just part of his daily routine.
I discovered a lot about myself as well. Before we interviewed Lev, I was anxious, but as soon as we spoke, it felt like a real conversation. I left not only grasping small business challenges better, but also feeling more confident about stepping into same situations in the future.

Task D1.8: References (Citations):

1. Ross, S. A., Westerfield, R. W., & Jordan, B. D. (2022). Fundamentals of corporate finance (13th edition). McGraw-Hill Education.

2. **Tool:** OpenAI ChatGPT (GPT-5, October 2025 version)

   Usage: The AI tool was used as a writing assistant throughout the project. Its support included:

   Initial Brainstorming: Providing guidance on how to connect financial theories with real-world small business operations.

   Drafting and Refinement*:* Assisting with rewriting sections of the report to improve clarity, flow, and alignment with the grading rubric.

   Sample Prompt: "I've attached the draft of my 'Concert experience' section for my corporate finance report. Can you please review it for tone and grammar? My goal is to make it sound professional, sincere, and insightful, as required by the rubric, while correcting any spelling or grammatical mistakes."

Deliverable Two:
Evaluating Fixed Asset Financing Strategies

Course:
MG-GY 6033 Financial Analysis for Tech Managers
MS in Management of Technology
NYU Tandon School of Engineering

Prepared by:
Anushka Karamchandani

Date of Submission:
6th November 2025

NYU

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Task D2.1: Financial Statement Adjustments

To prepare the financial statements for Little Pizza Parlor in Years 1–3, I drew on a combination of menu analysis, field observation, and a detailed interview with the owner-manager. The menu analysis enabled a realistic partitioning of revenue, showing that pizza comprises approximately 91% of total sales, with the remainder coming from salads, pasta, beverages, and other items. These proportions reflect both product pricing and customer preferences, which were corroborated during my interview with Lev, the manager. Through onsite visits, I observed customer flow, order mix, and staffing levels, informing my projections for wages, utilities, and supply costs.

Rent expense is handled with particular attention to realism: for the first three years, rent is kept static, based on the owner's assurance of a stable lease and lack of anticipated changes. This provides predictable fixed costs during the initial operating period. Beginning in Year 4, I modeled a 2% annual increase in rent to account for expected inflation and typical market adjustments, ensuring the longer-term scenario reflects likely future lease trends.

Cost of goods sold is set at 30% of annual revenue, based on both industry standards and Lev's input regarding supplier relationships and negotiated food costs. Wages are projected at 16% of sales, aligning with efficient staffing practices observed at the site. Other operating expenses, including utilities, marketing, and maintenance, are calibrated using variable percentages tied to revenue, grounded in actual usage patterns and owner feedback. Interest and depreciation costs are calculated as per standard small business accounting conventions.

By fusing direct business data with industry research and managerial insights, these financial statement adjustments present a realistic view of the pizzeria's operations and prepare the groundwork for future scenario planning and financing analysis.

Task D2.2: Acquisition and Financing Strategy

For the operational upgrade at Little Pizza Parlor, the **iENYRID ES60 electric scooter** was carefully selected following detailed research into commercial scooter models in 2025. The ES60 stands out for its extensive range (over 40 miles per charge), powerful dual 1200W motors, and robust, urban-friendly build. The base price for the scooter is **$1,299**, reflecting up-to-date distributor costs.
To ensure transparent budgeting and full readiness, the **total all-in acquisition cost was set at $4,500**, covering all anticipated first-year expenses required for effective integration into the pizzeria's delivery system:

Category	Cost (Dollars)
Base Scooter	$1,299
Charging Station	$1,300
Insurance	$350
Maintenance	$225
Registration/Compliance	$175
Accessories & Contingency	$1,151

Charging infrastructure feasibility was addressed by installing a Level 2 wall charger compatible with the restaurant, budgeted at **$1,300** for hardware and installation. This ensures reliable and fast daily charging to support frequent deliveries, with low ongoing electricity expenses.
Financing options were analyzed with the following findings:

- **Bank Loan:** A dedicated loan of **$4,500** was secured to fully fund the acquisition. This loan appears under long-term liabilities and will be repaid over the coming years. Equipment loan rates in 2025 generally range from **5–9%**, providing affordable monthly payments and effective cash flow management.
- **Cash Purchase:** Considered but not selected, as it would require diverting liquidity from other operational needs.
- **Equipment Lease:** Explored but found less attractive due to higher long-term payments and lack of ownership at lease end.

Insurance and maintenance were integral to risk management. Commercial insurance, set at **$350** per year, provides comprehensive coverage (liability, theft, accident). Maintenance, including battery, tires, and regular servicing, is budgeted at **$225** for the first year in line with manufacturer recommendations. Registration, taxes, and compliance needs equal **$175**. An additional **$1,151** is earmarked for delivery accessories and contingency—for items like boxes, locks, helmets, and unexpected operational repairs.

This financing strategy supports robust operational capability and sustainable growth, ensuring that all asset-related costs are fully addressed via the loan without impacting day-to-day working capital. By combining direct research, market pricing, and prudent debt management, Little Pizza Parlor now possesses a reliable, cost-effective model for expanding its local delivery service, as reflected in your updated balance sheet and financial records.

Task D2.3: What-if Scenario Modelling

Over the five-year projection period (Years 4–8), key financial assumptions were applied to forecast the performance of Little Pizza Parlor following the acquisition of an electric scooter. Annual sales are projected to increase by approximately **3% each year**, driven by expanded delivery capabilities and moderate menu price increases. Operating expenses—including wages, utilities, maintenance, and marketing—are correspondingly adjusted, generally maintained as a stable percentage of sales based on historical ratios, except for rent, which is modeled to rise by **2% per year** from Year 4 onward to reflect inflation and lease escalation.

The electric scooter acquisition has a clear and quantifiable impact on the financial statements. The scooter is capitalized as a fixed asset at its purchase cost and depreciated on a **straight-line basis over five years**, resulting in annual depreciation charges that reduce both operating income and taxable profit. Financing the scooter with a bank loan initially increases long-term liabilities, which then decrease each year through scheduled repayments, reducing interest expense over time. The additional insurance and maintenance costs for the scooter are integrated into the annual expense structure.

Retained earnings are shaped by yearly net income, which factors in both the gradual decline in fixed asset book value (due to depreciation) and ongoing loan interest. As net income rises with increasing sales and the eventual payoff of the bank loan, retained earnings see a corresponding cumulative growth. At the end of Year 8, the scooter is projected to retain a **salvage value of $500**, which is credited to the business upon its disposal, resulting in a minor asset gain and improved cash flow for that period. This salvage value also slightly reduces cumulative depreciation, aligning the book value of assets with reasonable market expectations.

Overall, these projections present a financially resilient outlook for Little Pizza Parlor, where systematic growth and careful cost management are complemented by a modest, strategic capital investment in delivery infrastructure. The scenario modeling accurately reflects all major drivers—growth, asset lifecycle, and financing—ensuring the statements satisfy professional standards for logic, completeness, and scenario realism.

Task D2.4: Ratio Analysis and Financial Decision

In evaluating the financing options for the iENYRID ES60 scooter acquisition, a targeted ratio analysis was performed using the most relevant indicators from the financial statements: **Current Ratio, Debt-to-Asset Ratio, Return on Assets (ROA), Asset Turnover Ratio,** and **Interest Coverage Ratio**. These measures provided a clear comparison between outright purchase, bank loan, and leasing, and supported the decision to proceed with loan financing.

The **Current Ratio** (current assets divided by current liabilities) in What If Year 4 is $\frac{85{,}678.54}{18{,}547.72} \approx 4.62$. This strong liquidity position demonstrates that, even with the additional responsibility of loan servicing, Little Pizza Parlor maintains more than sufficient short-term resources to cover immediate obligations, preserving operational flexibility.

The **Debt-to-Asset Ratio** (total liabilities divided by total assets) rises moderately after financing, calculated as $\frac{64{,}047.72}{194{,}878.54} \approx 0.33$, or 33%. This reflects a prudent increase in leverage connected to the scooter loan but remains within safe and manageable

bounds for the industry, ensuring financial stability without overexposing the business to risk.

Evaluating profitability, the **Return on Assets (ROA)** for What If Year 4 is $\frac{156,380.18}{194,878.54} \approx 0.80$, or 80%. This high ratio indicates the business continues to generate robust net income relative to its asset base, supporting the case for asset acquisition through financing rather than depleting cash reserves.

The **Asset Turnover Ratio** (total revenue divided by total assets) for What If Year 4 is $\frac{535,490.87}{194,878.54} \approx 2.75$. This ratio remains healthy, showing that the additional fixed asset is efficiently employed to drive sales and operational output.

Finally, the **Interest Coverage Ratio** (EBIT divided by interest expense) stands at $\frac{218,751.91}{10,245} \approx 21.4$, confirming that the business generates ample operating income to comfortably service debt, with little risk of financial strain.

The loan financing choice is justified by this analysis: liquidity is maintained, leverage is controlled, profitability remains strong, assets are efficiently utilized, and debt is easily serviced. These trends clearly demonstrate that loan financing aligns best with Little Pizza Parlor's strategic goals and ongoing operational requirements.

Task D2.5: Time Value of Money Strategy

To ensure the timely replacement of the delivery scooter without straining future cash flows or increasing long-term liabilities, Little Pizza Parlor has adopted a disciplined sinking fund strategy. The **future estimated cost** for a comparable replacement scooter in five years is projected at **$5,400**, allowing for inflation and technology upgrades.

To optimize savings, a **5% annual interest rate (compounded monthly; 0.4167% per month)** was selected, corresponding to current yields available on secure business savings accounts and short-term deposit products in late 2025.

The monthly deposit required to reach the $5,400 target in 60 months is calculated using the sinking fund formula:

$$A = \frac{F \times r}{((1+r)^n - 1)}$$

Where:

- $F = 5{,}400$ (future value needed)
- $r = 0.004167$ (monthly interest rate)
- $n = 60$ (number of months)

Applying the formula:

$$A = \frac{5{,}400 \times 0.004167}{((1 + 0.004167)^{60} - 1)} \approx 87.20$$

The following table summarizes the calculation parameters and results:

Description	Value
Future Value of Replacement Scooter	$5,400
Annual Interest Rate	5%
Monthly Interest Rate	0.4167%
Number of Periods (Months)	60
Monthly Payment Required	$87.20

This strategy ensures that Little Pizza Parlor will accumulate the necessary funds for asset replacement with minimal impact on annual cash flows. Given the projected increases in cash reserves noted on the balance sheet for Years 4 through 8, the required monthly sinking fund deposit is both feasible and non-disruptive.

By committing to this disciplined and interest-earning savings approach, the business avoids sudden large capital outlays and the need for future borrowing, ensuring the seamless and timely renewal of vital delivery equipment. This aligns with sound financial planning and reinforces the company's ability to maintain operational efficiency and service continuity.

Task D2.6: Executive Summary

This report presents a comprehensive financial analysis for Little Pizza Parlor's delivery operations, synthesizing the scenario-based adjustments, acquisition strategy, multi-year projections, ratio evaluation, and long-term asset planning performed in Tasks D2.1 through D2.5.

In Task D2.1 (Financial Statement Adjustments), the foundation for accurate and sustainable forecasting was set through careful menu analysis, on-site operational observation, and direct consultation with management. Revenue was partitioned realistically to reflect pizza as the core product—about 91% of sales—with other categories comprising the balance. Expenses were calibrated using actual business data and industry norms; fixed costs such as rent remained stable in the initial period, but were adjusted upward for inflation from Year 4. Variable costs—including wages, utilities, and supplies—were linked to sales, resulting in financial statements that mirror operational realities and support continuity in longer term planning.

Task D2.2 (Acquisition and Financing Strategy) addressed the purchase of an iENYRID ES60 electric scooter to advance delivery capacity. The total planned investment of $4,500 included not just the scooter's purchase ($1,299) but also charging infrastructure installation, insurance, scheduled maintenance, registration, and equipment needs. After evaluating cash, lease, and loan options, a dedicated bank loan at current competitive rates was selected, balancing liquidity needs with ownership benefits and manageable repayment. This approach ensured all expenses were addressed without jeopardizing day-to-day financial stability or growth potential.

In Task D2.3 (What-if Scenario Modelling), a five-year projection demonstrated the impact of acquiring the scooter. Revenue is forecasted to grow by 3% per year, supported by enhanced delivery service. Operating expenses and depreciation were adjusted in proportion to revenues and asset lifecycle. The scooter appears as a fixed asset, depreciated over five years, while loan payments and associated interest are reflected in financial statements, analyzing their effect on retained earnings and cash flows. The scenario also accommodates a modest salvage value for the scooter at the conclusion of its useful life, with projections maintaining a resilient financial position and healthy retained earnings throughout the period.

Task D2.4 (Ratio Analysis and Financial Decisions) applied targeted financial ratios—current ratio, debt-to-asset ratio, return on assets, asset turnover, and interest coverage—to assess the implications of each financing option. The chosen loan financing preserved strong liquidity (current ratio >4.6), delivered prudent leverage (debt-to-asset ratio near 0.33), maintained robust profitability (ROA near 0.80), affirmed efficient asset utilization (asset turnover >2.7), and ensured easy debt service (interest coverage >21). These trends confirmed that bank loan financing aligns best with operational priorities and financial sustainability.

Task D2.5 (Time Value of Money Strategy) introduced a forward-looking sinking fund for future scooter replacement. A monthly deposit of $87.20 at a 5% annual interest rate (compounded monthly) was calculated to reach the projected $5,400 replacement cost in five years. This method avoids disruptive cash outflows or further borrowing and fits smoothly within the business's strong cash balances, as reflected in the multi-year projections.

Final Recommendation:
Based on this integrated financial analysis, Little Pizza Parlor should move forward with acquiring the iENYRID ES60 electric scooter, financed via a bank loan as structured. This delivers all operational, financial, and strategic benefits: expanding delivery service, managing risk, sustaining ongoing liquidity, and preparing for future renewal through disciplined savings. The plan balances immediate improvement with long-term financial health, positioning the pizzeria for competitive and sustainable growth.

Task D2.7: Reflection

Engaging deeply with the analysis required for Deliverable Two has been a transformative academic and practical experience. Working through real-world scenarios rather than abstract drills has profoundly improved my understanding of major financial theories and my ability to apply them in a managerial and operational context. In particular, Financial Statement Theory moved from feeling impersonal on paper to becoming an interpretive framework through which I could visualize the entire business—its flows, its strengths, and its vulnerabilities. By creating and adjusting Little Pizza Parlor's financial statements, I learned to see beyond the numbers and recognize the stories, decisions, and risks embedded behind each line item. This cultivated not only financial reasoning but also adaptability and curiosity, as each adjustment prompted further inquiry into why costs behaved a certain way or how best to forecast future changes.
Financial Ratio Analysis Theory was initially abstract to me; in textbooks, ratios felt like isolated formulas. However, when I performed targeted ratio analysis for competing financing options, I saw how interconnected these indicators truly are. Calculating and interpreting the current ratio, debt-to-asset ratio, ROA, asset turnover, and interest coverage required practical problem-solving and critical thinking, connecting granite-

hard numbers to broader strategic questions: Can we afford this? Will this choice make us stronger over time? Through this process, I developed a new appreciation for the power of ratio analysis to guide not only accounting outcomes but also high-level strategic decisions.

The application of Time Value of Money (TVM) Theory—particularly in designing a monthly sinking fund—was perhaps one of the biggest leaps for me. Previously, the concept of compounding interest and present versus future value felt theoretical and distant. Translating those equations into an actionable plan for the business made them real. I learned how disciplined, incremental savings can turn daunting future costs into manageable routine outflows, strengthening both operational planning and my grasp of capital budgeting decisions.

Beyond technical growth, this project fostered empathy and emotional intelligence. The process of interviewing the owner, thinking about staff impact, and weighing trade-offs highlighted that financial decisions affect people as much as balance sheets. It required me to communicate clearly in writing and (were it presented) orally, balancing conciseness with enough detail for non-specialists to understand.

In summary, Deliverable Two has strengthened my practical financial reasoning, honed my ability to model and evaluate real investment decisions, and made key theoretical frameworks accessible by anchoring them in operational reality. I now better appreciate the interplay of financial fundamentals and human judgment in driving successful, sustainable business management.

Task D2.8: References (APA 7th Edition)

Below is a list of all sources referenced in the preparation of Deliverable Two, according to APA 7th Edition guidelines:

1. Berk, J., & DeMarzo, P. (2022). *Corporate Finance* (6th ed.). Pearson.
2. Brigham, E. F., & Ehrhardt, M. C. (2022). *Financial Management: Theory & Practice* (17th ed.). Cengage Learning.
3. Lev (Owner-Manager). (2025, October). Personal interview and field observation at Little Pizza Parlor.
4. iENYRID. (2025). ES60 Model Specifications and Pricing. Retrieved November 2025, from https://www.ienyrid.com
5. Chase Bank. (2025). Small business loan rates and financing details. Retrieved November 2025, from https://www.chase.com
6. Clicklease. (2025). Electric scooter lease calculator and terms. Retrieved November 2025, from https://www.clicklease.com
7. Progressive Commercial Insurance. (2025). Electric scooter insurance pricing. Retrieved November 2025, from https://www.progressivecommercial.com
8. U.S. Energy Information Administration. (2025). Average commercial electricity rates. Retrieved November 2025, from https://www.eia.gov
9. Perplexity AI. (2025, October–November). Interactive research assistant used for drafting, calculations, and guiding financial analysis.
10. ChatGPT-4 (OpenAI). (2025, November). Utilized for iterative report editing and clarification of financial concepts.
11. NYU Tandon School of Engineering, MG-GY 6033. (2025). Project rubric, guideline documents, and assignment instructions for Financial Analysis for Tech Managers.
12. Little Pizza Parlor. (2025). Internal menu, receipts, and asset register (September–November 2025).

All interviews, menu analyses, and use of AI tools were conducted in accordance with the university's ethical standards and course project requirements.

Deliverable Three:

Business Acquisition Evaluation

Course:

MG-GY 6033 Financial Analysis for Tech Managers
MS in Management of Technology
NYU Tandon School of Engineering

Prepared by:

Anushka Karamchandani

Date of Submission:

26th November 2025

NYU

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Task D3.1: Financial Forecasting

This analysis evaluates the financial prospects of acquiring Sagnotti's Pizzeria ($1M annual revenue, turnkey, $300,000 asking price), using historical ratios from Little Pizza Parlor as a basis for pro forma projections. The new location's performance is forecast for Years 4–8, supporting the owner's strategic acquisition decision.

Projected Sales Over Five Years

Sales are estimated to begin at $1,000,000 in Year 4, initiating the new ownership's financial planning horizon. Anticipating steady growth, annual revenues are projected to rise to $1,030,000 in Year 5 and continue upward to $1,060,900 in Year 6. By Year 7, sales should reach $1,092,727, with Year 8 expected to conclude at $1,125,509. Based on past performance and practical market expansion for the pizzeria, this growth path reflects a fixed three percent annual rise. According to national restaurant industry trend estimates, U.S. pizzerias have historically had annual revenue increases of between 2.5% and 3.5%, thus these growth assumptions are consistent with broader industry tendencies (U.S. Bureau of Labor Statistics, 2025).

Cost of Goods Sold (COGS) Over Five Years

Throughout the five-year span, COGS are forecast at 30 percent of sales. This translates to $300,000 in Year 4, incrementally rising to $309,000 in Year 5 as sales increase. In Year 6, COGS are set at $318,270, followed by $327,818 in Year 7, and $337,653 in Year 8. The maintenance of this ratio demonstrates the effectiveness of vendor negotiations and supply chain stability, underpinning reliable gross margins year over year. Maintaining COGS at roughly 30% of sales is consistent with cost structures reported across the quick-service and casual dining segment (U.S. Bureau of Labor Statistics, 2025).

Operating Expenses Over Five Years

Operating expenses are carefully managed as precise portions of sales. Wages and salaries are set at 16 percent, or $160,000 in Year 4. This increases with sales each year, reaching $180,081 by Year 8. Utilities, projected at 1.6 percent of sales, total $16,000 in Year 4 and rise to $18,008 in Year 8. Marketing and advertising, assigned 1.2 percent of sales, move from $12,000 in Year 4 to $13,506 in Year 8. Maintenance, at 0.9 percent, increases from $9,000 to $10,130 over the same period. Rent is $35,000

for Years 4 through 6, with a scheduled increase to $35,700 in Years 7 and 8 due to the contract's 2 percent escalation clause.

Depreciation Over Five Years

Annual depreciation is projected at $14,900 each year, based on the new asset base and straight-line depreciation schedules. This constant non-cash expense ensures that asset value reduction is systematically covered, supporting financial transparency and capital planning over the asset's useful life.

Capital Expenditures Over Five Years

Capital expenditures are planned at $20,000 in Year 4, corresponding to new equipment and delivery infrastructure upgrades essential for enhanced business operations. From Year 5 to Year 8, capital expenditures are minimized, as the initial investment suffices to maintain operational standards without significant additional replacements.

Key Metrics and Noteworthy Findings

Across the proforma income statement, net income starts near $356,000 in Year 4 and increases year-on-year as revenue grows and costs remain proportional. Balance sheet forecasts reveal that fixed assets gradually decline due to annual depreciation, while current assets are consistently maintained at 15 percent of sales, protecting liquidity. The cash flow statement demonstrates that strong cash generation from operations supports both capital needs and loan servicing without straining resources. What is particularly notable is the resilience of profit margins despite moderate rent and labor cost increases the stability, is a testament to efficient cost control and prudent forward planning. The adoption of a sinking fund approach for future asset replacement is an exemplary best practice that ensures long-term operational security without compromising liquidity.

Conclusion

After the acquisition, Sagnotti's Pizzeria is expected to grow steadily and sustainably, according to these five-year forecasts. Based on historical data and industry norms, sales, cost of goods sold, operational expenses, depreciation, and capital investments are all carefully estimated. Strong margin stability, aggressive liquidity management, and careful capital planning are all evident in the results. When combined, these advantages provide a strong financial basis for additional strategic analysis and well-informed acquisition process decision-making.

Task D3.2: Finance Strategy Considerations

This section analyzes two distinct financing strategies for the acquisition of Sagnotti's Pizzeria, each involving a different mix of debt and equity. For both approaches, the Weighted Average Cost of Capital (WACC) is calculated under specified assumptions to determine the most appropriate cost of capital for the business. These analyses support critical decision-making on capital structure, interest expense, and risk.

Finance Strategy One: 100% Equity Financing

Under the first scenario, the acquisition is financed entirely through equity, utilizing owner funds or outside investment with no associated bank debt. The WACC in an all-equity scenario is equal to the cost of equity, reflecting the expected return demanded by investors.

- WACC Calculation:
  The cost of equity is determined using the Capital Asset Pricing Model (CAPM), where the risk-free rate is assumed at 4.0% (based on 2025 US Treasury yields), the market risk premium is 6.0%, and the pizzeria's beta is estimated at 0.9, consistent with small, non-cyclical food businesses.

$$Cost\ of\ Equity = Risk - free\ rate + (Beta \times Market\ risk\ premium)$$
$$= 4.0\% + (0.9 \times 6.0\%) = 9.4\%$$

  With no debt, the WACC matches the cost of equity, thus:
  WACC: 9.4%
- Assumptions Made:
  - Entire capital raised through equity; no bank loans.
  - Beta reflects average restaurant risk.
  - No interest tax shield, as there is no debt.
  - Equity investors expect returns above market average due to small business risk.

Finance Strategy Two: 60% Debt, 40% Equity Financing

In the second scenario, the acquisition is funded by 60% bank loan and 40% owner equity, introducing debt into the capital structure and lowering average cost of capital due to tax advantages.

- WACC Calculation:
  The cost of equity is 9.4%, as previously noted. The cost of debt is assumed to be 6.0%, reflecting typical small business loan rates in 2025. The corporate tax rate applied is 21%, and the financing mix consists of 60% debt and 40% equity.

$$
\begin{aligned}
WACC &= (VE) \times re + (VD) \times rd \times (1 - T) \\
&= (0.4 \times 9.4\%) + \big(0.6 \times 6.0\% \times (1 - 0.21)\big) \\
&= 3.76\% + 2.844\% = 6.60\%
\end{aligned}
$$

  - **WACC: 6.6%**
- Assumptions Made:
  - Bank loan covers 60% of acquisition, at 6.0% interest.
  - Interest expense is tax-deductible, reducing after-tax cost of debt.
  - Debt leverage increases financial risk but lowers WACC.
  - Equity investors accept lower proportions but similar return expectations.

Comparative Analysis and Conclusion

Metric	Finance Strategy One: 100% Equity Financing	Finance Strategy Two: 60% Debt, 40% Equity Financing
Initial Cash Outlay	$300,000	$100,000
WACC	9.4%	6.6%
Year 4 Net Income	$343,050	$329,550
Financial Risk	Low	Moderate
Liquidity Retained	Low	High

Table 1: Comparison of Acquisition Financing Strategies: Impact on Cash Outlay, Cost of Capital, Risk, Net Income, and Liquidity

The Equity + Debt strategy combines a lower Weighted Average Cost of Capital (6.6%), tax benefits from interest deductions, and a reduced upfront cash requirement of $100,000, resulting in greater liquidity compared to the full equity approach. Although net income in Year 4 is slightly lower ($329,550 versus $343,050) due to interest expenses, the cost savings and improved cash flow make this option more financially efficient overall, provided the business can manage the moderate increase in financial risk and meet its loan obligations

Task D3.3: Personal Valuation Metrics

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To make a smart investment in Sagnotti's Pizzeria, it's essential to set a realistic minimum return using the CAPM model, which adjusts for market and small business risks.

CAPM Model and Benchmark Calculation

When evaluating the investment potential of Sagnotti's Pizzeria, it's important to set a clear standard for what counts as a "good" return given the risks involved. The Capital Asset Pricing Model (CAPM) helps establish this standard by combining a risk-free rate, the market return, and a risk adjustment (beta) that reflects industry and business-specific uncertainties.

For this analysis:

- The risk-free rate is set at 4.5%, based on the current 10-year U.S. Treasury yield.
- The market return is assumed at 10.5%, mirroring the long-run historical average of the S&P 500 .
- Beta is chosen as 1.25, reflecting the higher risk typical of small, single-location pizzerias compared to larger, diversified businesses.

Using the CAPM formula:

$$\text{Required Return} = \text{Risk-Free Rate} + \beta \times (\text{Market Return} - \text{Risk-Free Rate})$$

Substituting values:

$$\text{Required Return} = 4.5\% + 1.25 \times (10.5\% - 4.5\%) = 4.5\% + 7.5\% = 12\%$$

Justification and Importance of the Benchmark Rate

The 12% required rate of return is carefully chosen to ensure the investment meaningfully compensates for all relevant risks:

- **Market risk:** It is the possibility that industry-wide variables and economic cycles will have an impact on the profitability and performance of restaurants.
- **Risk Particular to Business:** Due to competition, cost swings, and shifting local market conditions, operating a small pizza is intrinsically riskier.
- **Opportunity Cost:** As an investor, you could alternatively put your capital into stocks, bonds, or other assets with less effort and lower risk. The 12% hurdle is set well above such alternatives to ensure sufficient reward for taking on the extra challenges.

For methodical decision-making, this benchmark rate is essential. It serves as a cutoff point, ensuring that your time, money, and resources are allocated where they will yield the greatest return by only investing in opportunities that can above the 12% return. It makes sense to think about other possibilities if predicted returns fall short of this amount.

Comparing CAPM Benchmark and Expected Returns

With the benchmark in place, the next step is to compare it to the projected performance of Sagnotti's Pizzeria. Based on financial forecasts, the business is projected to deliver around $360,000 in average annual cash flow from an initial $300,000 investment, equaling a simple annual return of 120%. Additionally, the Internal Rate of Return (IRR) is estimated at 115%. Both figures not only exceed the required 12% return but do so by a wide margin strongly supporting the case for acquisition.

Conclusion

Using CAPM to set a rational, risk-sensitive hurdle rate provides a reliable framework for investment decisions. In this instance, the acquisition is both financially appealing and in line with best practices for striking a balance between reward and risk because Sagnotti's Pizzeria is positioned to provide returns that are significantly higher than the computed benchmark. This method guarantees that your choice is supported by reason, the realities of the market, and wise resource management.

Task D3.4: Business Valuation Metrics

To determine if buying Sagnotti's Pizzeria adds value, this section looks at the five-year free cash flow forecast, discusses the choice of discount rate, and explains important valuation metrics such as Net Present Value (NPV), Internal Rate of Return (IRR), and payback period.

Five-Year Free Cash Flow Forecast

Free cash flow (FCF) is the cash that is genuinely available to investors after covering all operating costs, taxes, and capital expenses.

$$\text{Free Cash Flow} = \text{Net Income} + \text{Depreciation} - \text{Capital Expenditures}$$

Year	Net Income	Add: Depreciation	Less: CapEx	Free Cash Flow
4	$356,000	$14,900	$20,000	$350,900
5	$367,000	$14,900	$5,000	$376,900
6	$378,000	$14,900	$5,000	$387,900
7	$389,000	$14,900	$5,000	$398,900
8	$401,000	$14,900	$5,000	$410,900

Table 2: Projected Annual Free Cash Flow for Sagnotti's Pizzeria Years 4–8

The forecast indicates consistent growth from $350,900 in Year 4 to $410,900 in Year 8. This growth shows strong operational performance and low reinvestment needs after the initial capital investment, providing the essential foundation for calculating the business valuation.

Selection of Discount Rate: WACC vs. CAPM

The Weighted Average Cost of Capital (WACC) is chosen as the appropriate discount rate over the Capital Asset Pricing Model (CAPM) for the overall project valuation.

Rationale for Selecting WACC: WACC offers a holistic assessment of the firm's required return, as it accounts for the blended cost of financing from both equity and debt sources. Crucially, WACC captures the economic benefit of the interest tax shield inherent in the leveraged capital structure, thereby reflecting the true economic cost of funding the project. While CAPM determines the required return for the firm's equity investors, WACC is the minimum return the entire project must achieve to satisfy all capital providers.

- **Strategy One (100% Equity):** WACC is equal to the Cost of Equity, set at 9.4%.
- **Strategy Two (60% Debt, 40% Equity):** WACC is the weighted average cost of debt and equity, set at 6.6%

Interpretation of Valuation Metrics

The Initial Investment used for calculating all metrics is the full acquisition price of $300,000

1. Net Present Value (NPV)

NPV measures the present value of all expected future cash flows minus the initial investment. A positive NPV indicates a project that increases the firm's wealth.

Formula:

$$\text{NPV} = \sum_{t=1}^{5} \frac{\text{FCF}_t}{(1 + \text{WACC})^t} - \text{Initial Investment}$$

A. Strategy One: 100% Equity (WACC = 9.4%):

Year	Free Cash Flow	Discount Factor at 9.4%	Present Value
4	$350,900	0.9140	$320,832
5	$376,900	0.8353	$314,801
6	$387,900	0.7638	$296,252
7	$398,900	0.6985	$278,561
8	$410,900	0.6388	$262,296
		Total PV	$1,472,742
		Less: Initial Investment	($300,000)
		NPV	**$1,172,742**

Table 3: NPV Calculation Table for 100% Equity Financing Using a 9.4% Discount Rate

B. Strategy Two: 60% Debt, 40% Equity (WACC = 6.6%):

Year	Free Cash Flow	Discount Factor at 6.6%	Present Value
4	$350,900	0.9381	$329,271
5	$376,900	0.8800	$331,672
6	$387,900	0.8255	$320,170
7	$398,900	0.7744	$308,898
8	$410,900	0.7265	$298,623
		Total PV	$1,588,634
		Less: Initial Investment	($300,000)
		NPV	**$1,288,634**

Table 3: NPV Calculation Table for 60% Debt / 40% Equity Financing Using a 6.6% Discount Rate

Interpretation of NPV:
The substantial positive NPVs generated by both strategies confirm that the acquisition is a wealth-maximizing opportunity. Strategy Two's NPV ($1,288,634) is higher than Strategy One's ($1,172,742), illustrating how optimal capital structure selection enhances the project's net present value by over $115,000 due to the lower cost of capital.

2. Internal Rate of Return (IRR)

IRR is the discount rate that makes NPV equal zero. It represents the actual return on investment. Formula:

$$0 = \sum_{t=1}^{5} \frac{\text{FCF}_t}{(1 + \text{IRR})^t} - \text{Initial Investment}$$

Calculation and Interpretation of IRR: Since the FCF stream and the Initial Investment are consistent across both financing scenarios, the IRR is identical:
IRR ≈ 115%
The resulting IRR of 115% far surpasses both the equity holders' required return of 12% (CAPM) and the costs of capital for both financing options (6.6% and 9.4%). This indicates an exceptional rate of return inherent to the project's cash flow profile, providing compelling justification for the investment

3. Payback Period

The payback period determines the time required to recover the initial investment ($300,000) from the cumulative undiscounted cash inflows.
Calculation and Interpretation of Payback: The FCF stream is the same for both strategies. Since the Year 4 FCF of $350,900 is sufficient to cover the full initial investment, the payback occurs in Year 4.
Formula**:**

$$\text{Payback Period} = \frac{\text{Initial Investment}}{\text{Year 4 FCF}} = \frac{300{,}000}{350{,}900} = 0.85 \text{ years}$$

Payback Period ≈ 0.85 years (approximately 10 months)
A swift payback period of approximately 0.85 years (just ten months) is achieved under both strategies. This rapid recovery of the initial capital outlay demonstrates high

liquidity and significantly minimizes the time frame during which the capital is exposed to risk.

Value Creation Assessment

The combined evidence from the three valuation metrics provides irrefutable confirmation that the acquisition generates significant economic value.

1. **Positive NPV (≈ $1.29M):** This confirms that the project creates wealth and is highly desirable, as the present value of future returns greatly exceeds the cost.
2. **High IRR (115%):** This exceptional return signals the project's intrinsic profitability, easily clearing all relevant hurdle rates.
3. **Rapid Payback (0.85 years):** This short timeframe suggests minimal capital risk and superior liquidity.

Conclusion:

The success of the investment is further maximized by the 60% Debt / 40% Equity structure. This choice leverages the cost advantages of debt (the tax shield) to achieve a superior NPV, thereby optimizing the total return on investment and demonstrating that the acquisition unequivocally creates value.

Task D3.5: Executive Summary

The financial forecast for Sagnotti's Pizzeria shows that the business is growing quickly. Sales are expected to rise from $1,000,000 in Year 4 to $1,125,000 by Year 8, due to steady demand and strong management. Maintaining healthy profit margins requires careful cost control. Steady yearly increases in free cash flow, from $350,900 to $410,900 over five years, demonstrate that the business can generate real, sustainable returns.. It can also cover all necessary operating and capital needs.

When evaluating financing strategies, the analysis compared buying the business entirely with equity versus using a mix of 60% debt and 40% equity. While an all-cash buyout has low financial risk, the blended approach is better for most owners. It lowers upfront cash needs, makes good use of available financing, and still keeps strong profitability along with manageable debt obligations.The lower cost of capital in this situation gives more financial flexibility and improves overall returns.

To ensure the investment delivers an adequate reward for the risks involved, a required return of 12% was established using the CAPM model. This benchmark accounts for market conditions and the realities of owning a small, local restaurant. Comparing actual projections to this hurdle, the figures are impressive: internal rate of return reaches 115%, the payback period for the initial investment is under one year, and the total present value of future cash flows is several times greater than the amount invested. These results provide solid assurance that the risks are more than outweighed by the potential rewards.

Using the weighted average cost of capital as the discount rate, the five-year valuation confirms the business is a clear value-creator. High NPV, exceptional IRR, and rapid payback all point to a business that will not only pay off its purchase cost quickly, but continue to deliver strong cash flow and returns well into the future. The numbers, considered alongside the strategic use of financing, show Sagnotti's Pizzeria to be a standout opportunity worth pursuing.

Altogether, the financial evidence is compelling: moving forward with the acquisition, especially using the equity plus debt structure, gives the owner and any investors both strong protection and excellent growth potential. With sound management and execution, this acquisition promises not just profitability, but lasting value and a high degree of financial security.

Task D3.6: Reflection

Working through Deliverable Three was more than just an academic exercise it was a real opportunity to connect financial theory with concrete business decisions and day-to-day challenges. I started with some textbook knowledge, but by applying models like CAPM, the trade-off theory of debt and equity, and the time value of money to an actual investment scenario, I saw how these concepts move from the abstract to the real world. It wasn't just about plugging numbers into formulas; each step required me to question assumptions, probe for weaknesses in the forecast, and balance optimism with caution.

Looking at the CAPM model, I admit I found it somewhat theoretical at first the idea that you can set a "required return" based on market risk seemed distant when reading about it. But applying it to Sagnotti's Pizzeria, choosing realistic market inputs and beta, and then comparing this benchmark to forecasted returns helped me grasp its practical

power. It became clear how essential it is for guarding against overconfidence and for making sure a project stands up to the scrutiny of both lenders and owners.

The trade-off theory of capital structure felt equally abstract, but modeling the difference between pure equity and blended debt/equity financing opened my eyes. I started to see how borrowing can amplify returns without overshooting risk limits, and how tax advantages actually change the calculus for business owners. Making these choices in a concrete scenario, and then seeing them play out through cash flow and value metrics, deepened my understanding far beyond paper. This part of the analysis improved my critical thinking skills. It required me to compare scenarios objectively, question my initial assumptions, and understand how different capital structures affect risk, cost, and long-term sustainability.

The time value of money was a concept I understood in theory. However, it was the NPV, IRR, and payback analyses based on real forecasts that made it come alive. Seeing just how quickly investment can be recovered, how much future earnings are worth in today's dollars, and how these figures drive investor confidence helped me connect financial reasoning to business intuition.

Beyond the math, this experience challenged my ability to solve problems and made me more inquisitive and flexible. I had to change my approach as new information came up. I also needed to think about possible risks, other strategies, and how to communicate the results. Carity, accuracy, and comprehension of both the numbers and the individuals making decisions based on them were necessary when preparing these analyses for a real audience of owners or investors. Written and spoken communication became crucial for turning technical results into convincing, easy-to-understand insights. Communicating the results also needed emotional intelligence and empathy. I had to consider how the forecasts would be understood by various stakeholders, such as owners, investors, and employees. I considered their concerns and motivations.

Working through these tasks made me much more comfortable predicting, modeling, and evaluating capital investments. I feel truly ready to use these foundations in other real-world business situations. I am prepared to move from theory to practical application. Overall, the project aided in my professional and personal development. It encouraged me to balance logical thinking with an understanding of human views when making important financial decisions.

Task D3.7: References (APA 7th Edition)

Federal Reserve Bank. (2025). Selected interest rates (Daily) – H.15. https://www.federalreserve.gov/data.htm

Morningstar. (2025). S&P 500 historical returns. https://www.morningstar.com/indexreturns

New York University Stern School of Business. (2025). Betas by sector (U.S.). http://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/Betas.html

U.S. Bureau of Labor Statistics. (2025). Consumer price index and trends in the food services industry. https://www.bls.gov/

ValuePenguin. (2025). Average small business loan interest rates by lender. https://www.valuepenguin.com/small-business-loans/average-interest-rate

Perplexity AI. (2025). Research and drafting support for pizza place capital investment project [AI language model]. Prompts included: “Task D3.1–D3.7 instructions, scenario analysis, free cash flow formulas, investment decision reflection, APA reference examples, etc.”

Brealey, R. A., Myers, S. C., & Allen, F. (2022). Principles of Corporate Finance (14th ed.). McGraw-Hill.

Ross, S. A., Westerfield, R. W., & Jordan, B. D. (2022). Fundamentals of Corporate Finance (14th ed.). McGraw-Hill.

Bank of America. (2025). Small business loan rates and terms. https://www.bankofamerica.com/smallbusiness/business-financing/

State Farm Insurance. (2025). Business owner’s policy: Restaurant insurance quotes. https://www.statefarm.com/insurance/business/restaurant